

OTEPIC

COMPOST MAKING



Otepic Kenya

P.O. Box, 4627 - 30200

Kitale, Kenya

+254 725 429 179

otepic07@yahoo.com

www.otepic.com

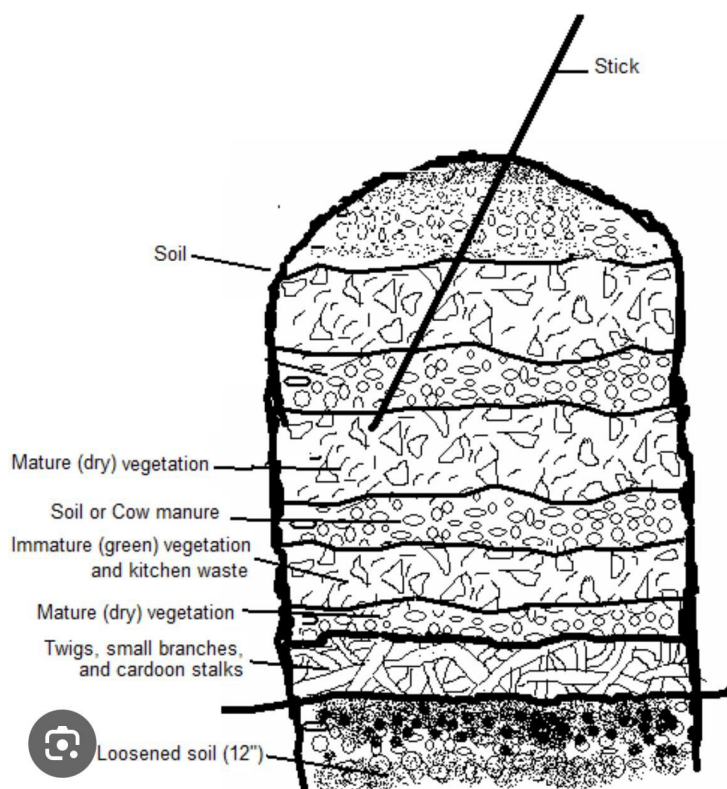
TRAINING MANUAL HANDBOOK

COMPOST MAKING

For people who are used to depending on the supermarket for food, it is easy to forget that what we eat depends entirely on fertile soil, Sir Albert Howard, a pioneer of the organic agriculture movement, considered that the fertility of the soil determines the future of civilization. The North African desert for example, used to be the grain growing area for Rome - until it was strip mined of nutrients by improper farming practices.

Nature manages the fertility of soil quite effectively with her natural cycles of life and death, growth and decay. Plants and animals live and grow their leaves, roots and residues enriching the soil of their environment. When they die, moist soil decomposes their bodies and transforms them into organic matter that replenishes and promotes new life and growth. Nature is an expert at recycling all of her wastes -- “living” recycling – so that organic materials, major minerals and trace minerals are continually being returned to the soil to nourish new growth.

COMPOST PREPARATION LAYOUT



THE BENEFITS OF COMPOST

1. **Improved Soil Structure:** It breaks up clay and clods and binds together sandy soils. It helps proper aeration clay like and sandy soil possible.
 2. **Moisture Retention:** It holds 6 times its own weight in water. A soil with good organic matter content soaks up rain like a sponge and regulates the supply to plants. A soil stripped of organic matter resist water penetration,
 3. **Aeration:** Plants can obtain 96% of these nutrients they need from air, water, and the sun (through the process of photosynthesis). A loose, health soil assists in diffusing air and moisture into the soil and exchanging nutrients.
 4. **Fertilization:** Compost is much better for the soil than chemical fertilizers, which do not add organic matter and some of which can leach out of the soil if the plants do not use them immediately. Compost contains both macro and micro nutrients, which are essential to plant growth as opposed to chemical fertilizers, which only contains specific soil nutrients such as DAP or CAN.
 5. **Nitrogen Storage:** the compost pile is a storehouse for nitrogen. Because it is tied in a compost break down process, water soluble nitrogen does not leach out or oxidize into the air a period of 3-6 months or more depending how the pile is constructed and maintained.
 6. **pH buffer:** A good percentage of compost in the soil allows plants to grow better in less-than-optimal pH situation.
 7. **Soil Toxin Neutralizer:** Important recent studies shows that plants grown in organically composted soils take up less lead, heavy materials and other urban pollutants.
 8. **Nutrient Release:** Organic acids dissolve soil minerals and make them available to plants. As the organic matter decomposes it releases nutrients for plants uptake and for the soil microbial population.
- Food for Microbial Life:** Good compost creates health conditions for soil organisms. Compost harbors earthworms and beneficial fungi that fight nematodes and other soil pest.
- Ultimate in Recycling:** The earth provides us with food, clothing and shelter. And we also

close the cycle in offering fertility, health, and life through composting. The pile recycles garden debris, leaves, and kitchen waste into food for the soil.

Other benefits of compost include those listed below.

- Prevents Soil Erosion
- Assists in Stormwater Management
- Promotes Healthier Plant Growth.
- Conserves Water.
- Reduces Waste.
- Combats Climate Change.
- Reduces Project Maintenance Costs.
- Improves Soil Health.

THE DECOMPOSITION PROCESS

The decomposition process that goes on in the compost pile is carried out by a succession of microscopic organisms including bacteria and fungi, and larger organisms, including earthworms. Providing ideal conditions for these organisms is what makes a good compost pile.

The compost pile needs:

Air – Beneficial bacteria need air to breathe, so compost materials should be piled up loosely, but not too loosely – too much air is not good, either.

Moisture – Soil organisms need enough water to keep them alive, but not too much – you do not want to drown them! The pile should be wet, but not too wet – like a well-wrung-out sponge.

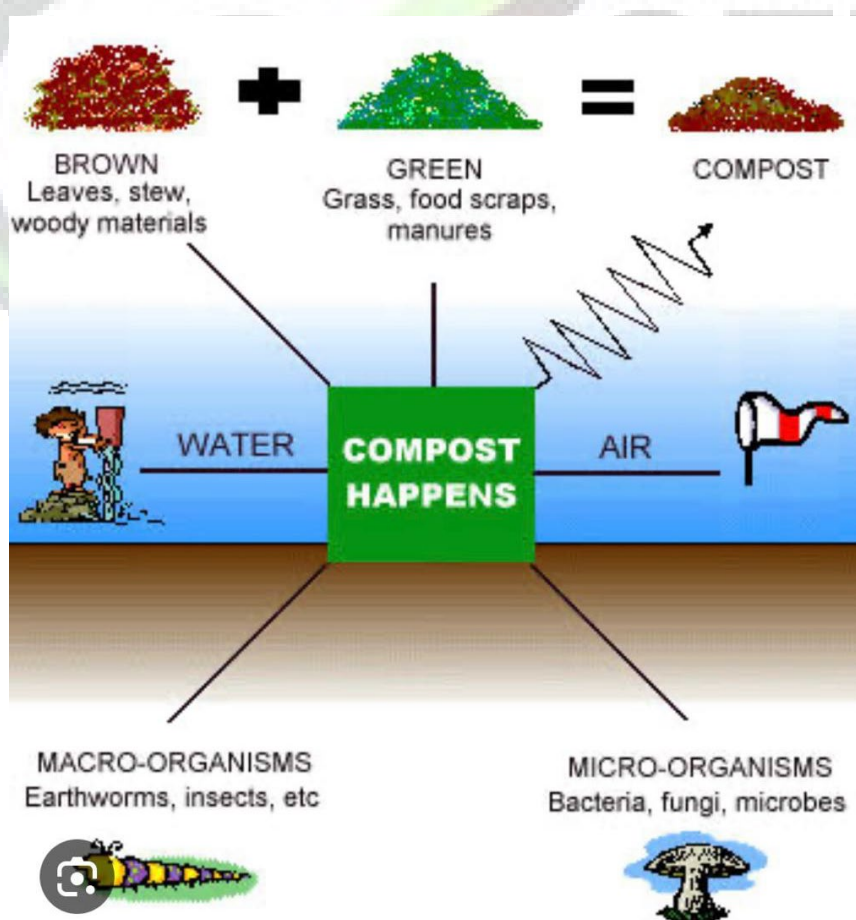
A variety of materials – The greater the variety of materials in a pile, the greater the variety of microbial life, and therefore the higher the quality of the compost and soil. In addition, greater microbial diversity reduces the likelihood of plant diseases.

Warmth – Microorganisms are most active during the warmer months of the year when the rate of decomposition is greater. It is important to build compost any time you have materials,

however, even when the weather is cooler and decomposition is slower.

As the decomposition process begins, the activity of the microorganisms will cause the pile to heat up. Some microorganisms will die and others will take over, continuing the process. Eventually, the soil organisms will change the original organic materials into a more stable form of organic matter called humus. Humus is a living fertilizer, alive with microorganisms consuming other microorganisms that have broken down, recombined and transformed the original organic matter. The nutrients in the humus are easily available to the plants in a slow, natural, continual process. What wonderful gardens there will be when we all develop a better sense of humus.

The Cycle



MATERIALS FOR A COMPOST PILE

The compost pile needs three kinds of materials:

1. **Dry vegetation**—Dry grass and weeds, leaves, straw, hay, dry compost crops, including some woody materials, such as broken-up corn stalks. Dry material provides carbon that is the energy source for all life forms.



2. **Green vegetation** – fresh weeds, green grass, kitchen wastes including a small amount of bones (but not meat – you do not need dogs and raccoons digging through your compost pile – or large amounts of oil), green compost crops. Green material provides nitrogen that enables the microorganisms to develop their bodies which are necessary to digest their carbon energy source.



3. Soil – good bed soil with valuable microorganisms to start the decomposition process. The soil will keep down flies and odors help the pile to hold water, and therefore allow the pile to decompose more slowly, which will ensure an easier-to-maintain compost pile. Some people like to keep their compost under control in a bin and a few uses a drum or an enclosure of some kind. This is not necessary, but if you prefer to use some sort of container, make sure the compost has enough air to breathe, so that the correct kind of decomposition can take place.



The following materials should not be included in the compost pile:

- Plants infected with disease or a severe insect attack, where eggs could be preserved or where the insect themselves could survive in spite the heat from the compost.
- Poisonous plants such as hemlock, which harm the soil.
- Plants that take too long break down such as the leaves grevillea robust
- Plants that have toxic to other plants and microbial life such as eucalyptus, cypress and whistling pine
- Plants that may be too acidic or contain substances that interfere the decomposing process such as spine needles, which are extremely acidic and contain a form of kerosene (however, especially compost pile are often made of acid materials, such as pine needles and leaves.

BUILDING A COMPOST PILE

When building your pile think about the layers that make up a dinner casserole like lasagna.

1. With a spading fork, loosen the soil 12 inches deep where you will build your pile.



2. Put down a 3-inch layer of rough materials which can help aerate the pile: twigs, small branches, corn or sunflower stalks, cane berry or rosebush pruning, and so on.



3. Make your compost “lasagna” layers, watering each layer as you go:
 - a. A 2” layer of dry materials.
 - b. A 2 layer of green materials.



4. A layer of soil which lightly covers the materials, or about $\frac{1}{2}$ of a 5-gallon bucket for a 3-foot by 3-foot compost pile.
5. Continue to add layers until your pile is about 3-feet high. If your pile is bigger at the base, you can pile the materials 4-feet high or more (but watch that the top of the pile does not start to slide). You can use a pitchfork to pull out the sides of the pile as you add layers, to keep the pile square.
6. Cover the top of the pile with extra soil, to maintain the moisture in the pile. A light layer of straw on top of the soil during the rainy season will keep out excess moisture and will prevent the pile from becoming soggy.

7. Water the pile as needed to keep it moist. Check the moisture in the middle of the pile from time to time—it is easy to either underwater or over water the pile.



8. Turn the pile after about 3-6 weeks. The purpose of turning is to bring the drier, less decomposed material on the outside. A good tool to use is a pitchfork, since it is lighter than a spading fork and is shaped to allow easy turning of the material. Start by loosening the soil in an area about one half to two thirds the original area (since the pile has shrunk), and add a layer of rough materials at the bottom. Move the materials from the original pile to the new pile, bringing the drier materials to the inside. Add water as you go, if necessary, to be sure that the turned pile is evenly moist.
9. Let the pile decompose- or “cook”, for a total of 3-6 months. The compost is ready to use when:
 - a. Most of the original ingredients are unrecognizable.
 - b. Its smell is fresh and woody like fresh spring water and
 - c. The material is dark brown or black, soft and crumbly
10. If you are not ready to use the compost when it is cured, spread it out and let it dry. It is important not to let the pile decompose too long, or the materials will turn into topsoil rather than compost and will lose the high-quality organic matter that has been so carefully built up.

This recipe may make building compost sound more complicated than it is. The important thing is just to go ahead and build your compost pile as the materials become available, without worrying too much about the details at the beginning. We recommend this low maintenance approach, since it is easy and produces such a good result.

The three most important elements in building compost are:

1. Have enough air in the pile.
2. Use as many different compost materials as you can and
3. Keep the compost moist enough.

As you learn to better understand the composting process and as you garden produces more and more material for you to use. You will be able to improve your technique.

USING YOUR COMPOST

The best time to put compost in your growing beds is the spring, just before transplanting the seedlings for the growing season. As a general rule, you can spread ½ inch of cured compost over the surface of the bed.

This comes to approximately six 5-gallon buckets per 100-square-foot bed. Then work it evenly in to the top 2 inches of the soil, using a one application of compost per 4-month growing season is adequate.

